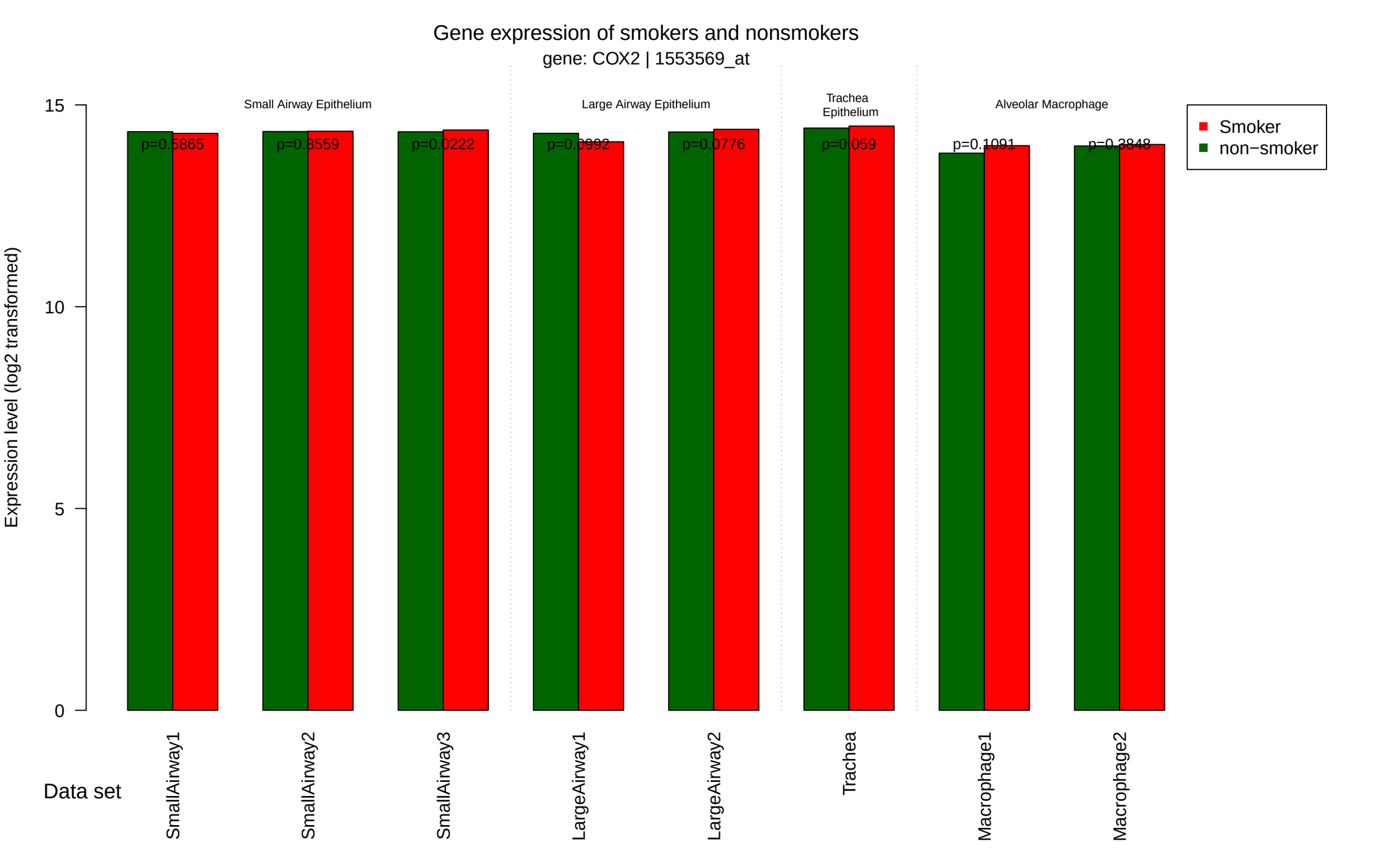
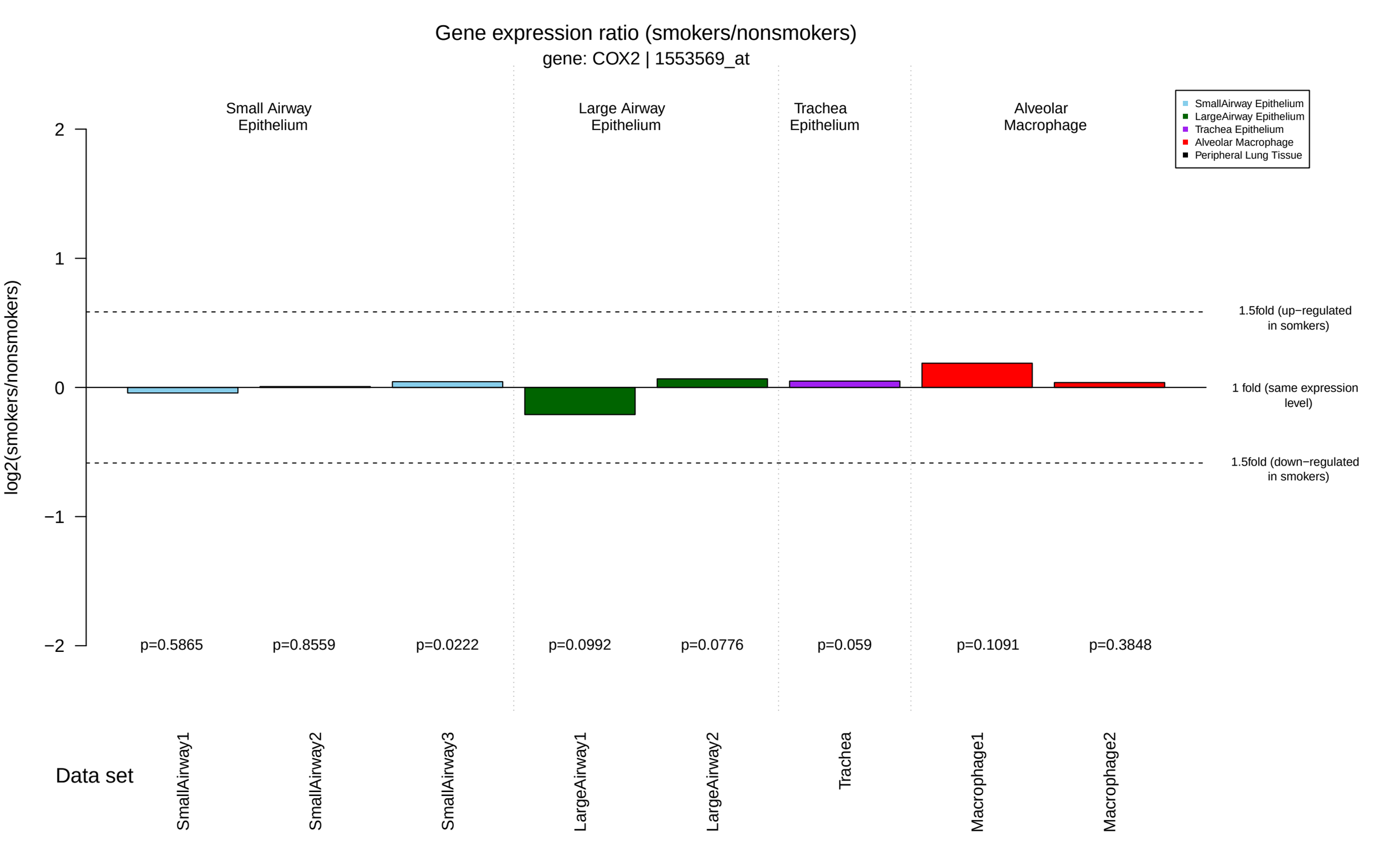
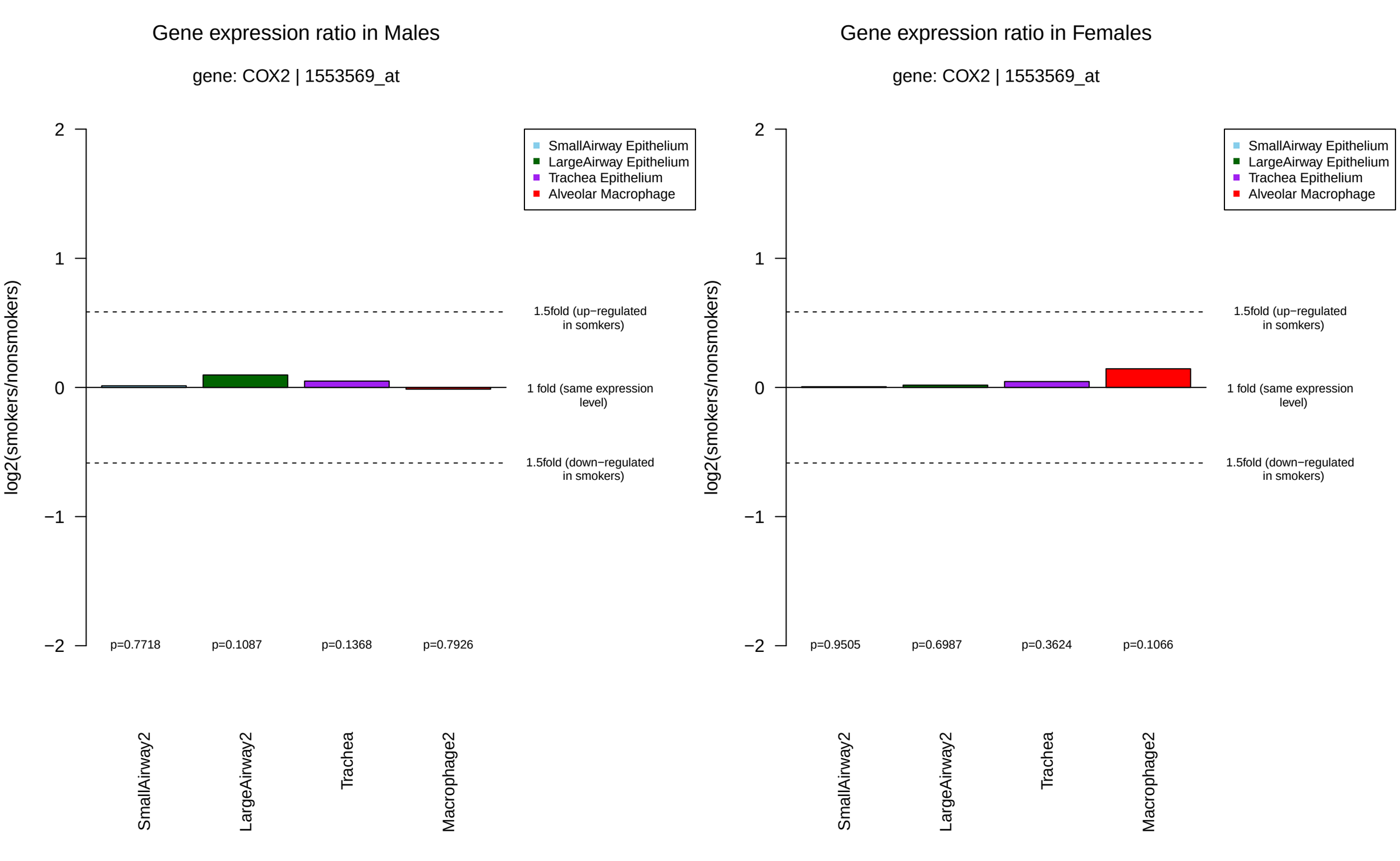
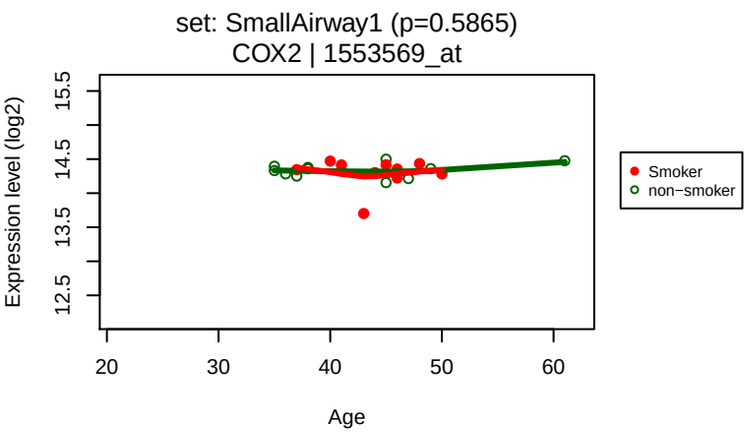
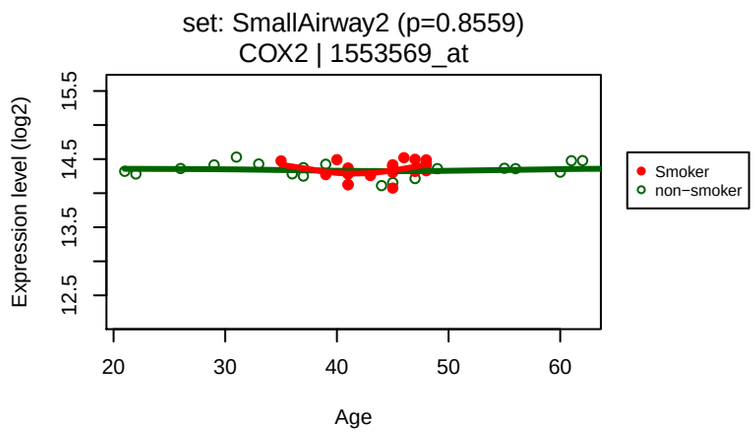
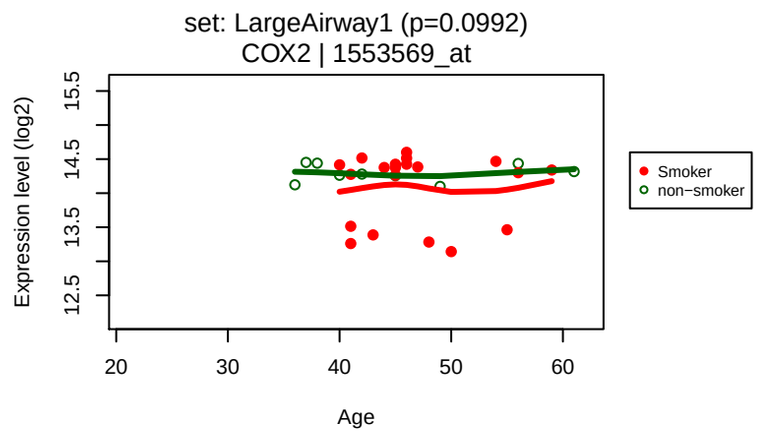

Gene expression of smokers and nonsmokers
gene: COX2 | 1553569_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5865
p=0.8559
p=0.0222
p=0.0992
p=0.0776
p=0.059
p=0.1091
p=0.3848
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

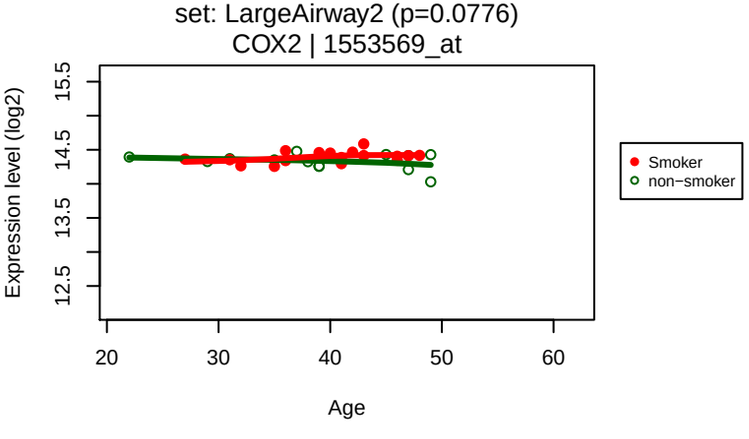

Gene expression ratio (smokers/nonsmokers)
gene: COX2 | 1553569_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5865
p=0.8559
p=0.0222
p=0.0992
p=0.0776
p=0.059
p=0.1091
p=0.3848
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

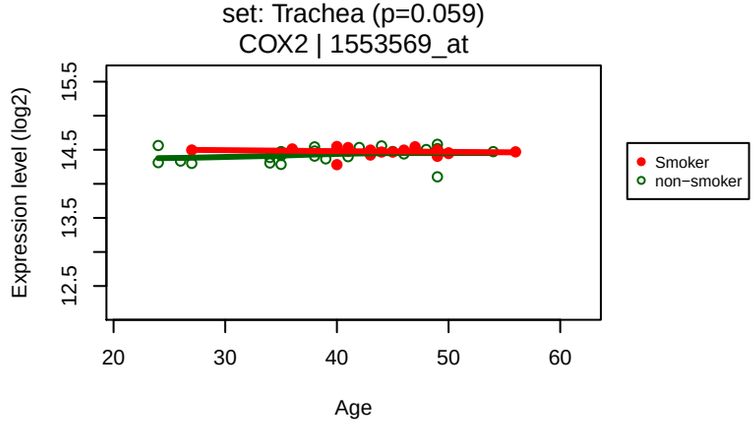

Gene expression ratio in Males
gene: COX2 | 1553569_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7718
p=0.1087
p=0.1368
p=0.7926
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COX2 | 1553569_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9505
p=0.6987
p=0.3624
p=0.1066
SmallAirway2
LargeAirway2
Trachea
Macrophage2

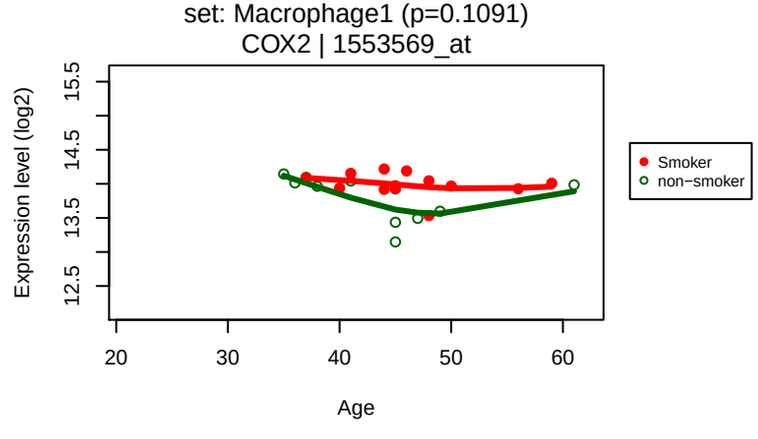

set: SmallAirway1 (p=0.5865)
COX2 | 1553569_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker

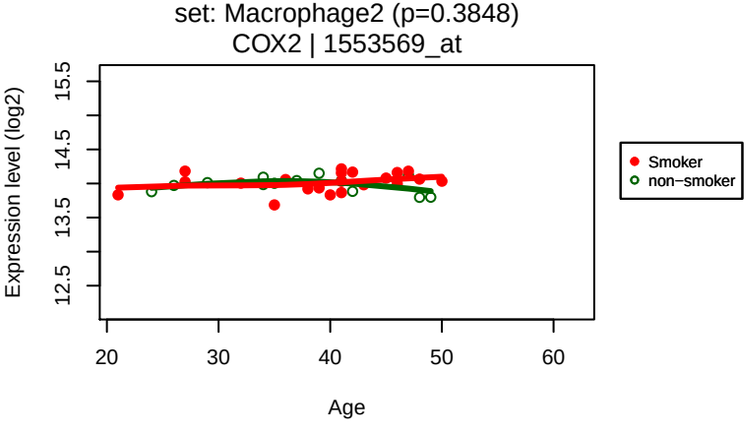

set: SmallAirway2 (p=0.8559)
COX2 | 1553569_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker

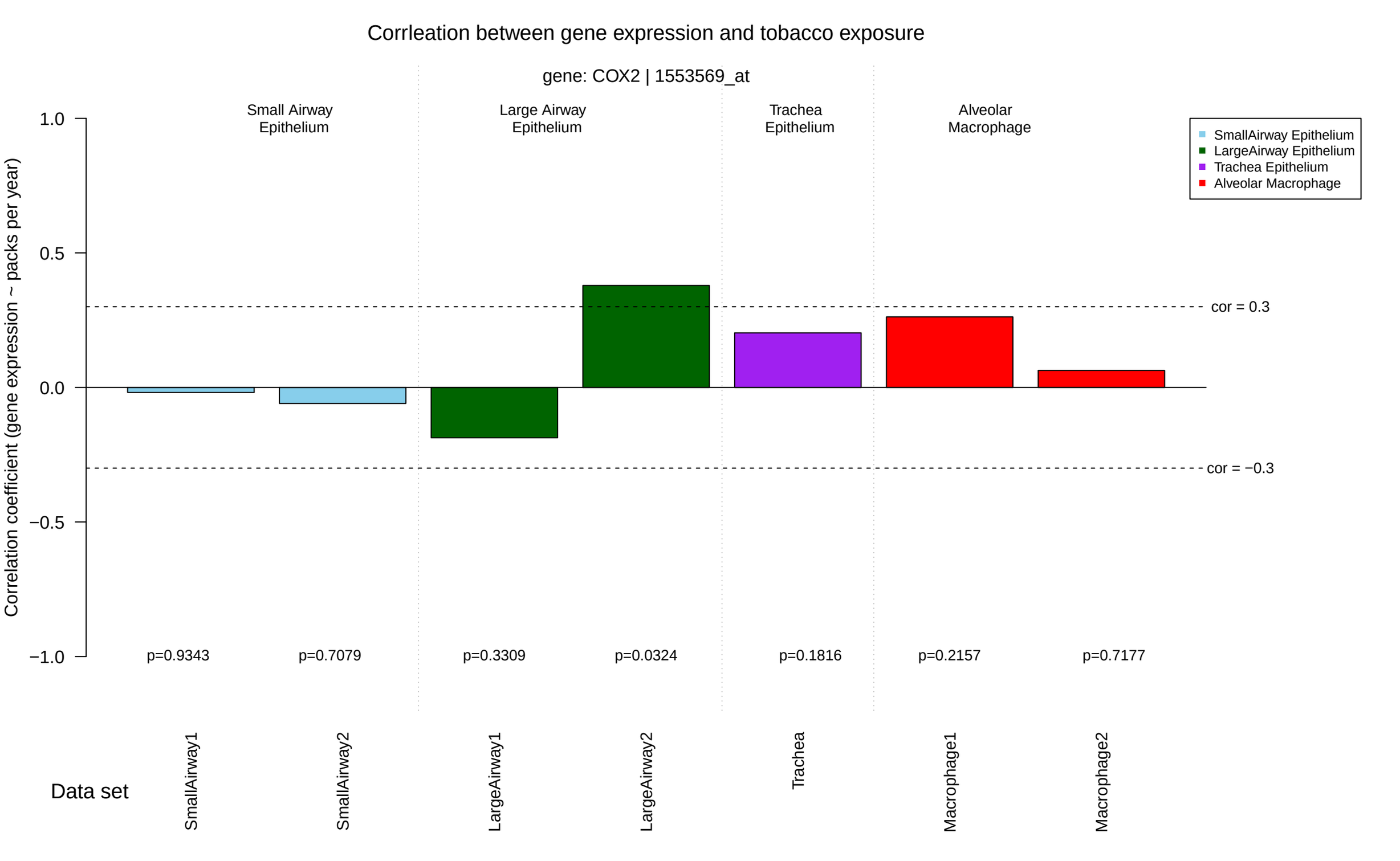

set: LargeAirway1 (p=0.0992)
COX2 | 1553569_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0776)
COX2 | 1553569_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.059)
COX2 | 1553569_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1091)
COX2 | 1553569_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3848)
COX2 | 1553569_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COX2 | 1553569_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9343
p=0.7079
p=0.3309
p=0.0324
p=0.1816
p=0.2157
p=0.7177
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2